PRACTICALNO.01

1] Practice Lab Assignments:

i) To accept an object mass in kilograms and velocity in meters per second and display its momentum. Momentum is calculated as e=mc2 where m is the mass of the object and c is its velocity.

=> # Program to calculate momentum

```
mass = float(input("Enter mass in kg: "))

velocity = float(input("Enter velocity in m/s: "))

momentum = mass * velocity

print("Momentum =", momentum, "kg m/s")
```

```
>>>
    ================== RESTART: C:/Users/Student/Desktop/IT/1.py =============
    Enter mass in kg: 70
    Enter velocity in m/s: 60
    Momentum = 4200.0 kg m/s
>>>
```

ii] Write a Python program for following conditions.

If n is single digit print square of it.

If n is two digit print square root of it.

If n is three digit print cube root of it.

```
import math
n = int(input("ENTER A NUMBER:"))
if 0 <= n<10:
   print(f"Square of {n}: {n**2}")
elif 10 <= n <100:
   print(f"Square root of {n}: {math.sqrt(n):.2f}")
elif 100 <= n <1000:
   print(f"Cube root of {n}: {n**(1/3):.2f}")
else:
   print("PLEASE ENTER A NUMBER BETWEEN 0 AND 999. ")
```

```
>>>
================== RESTART: C:/Users/Student/Desktop/IT/2.py ==================
ENTER A NUMBER:400
Cube root of 400: 7.37
>>>
```

III] Read the birthdate and salary in rupees of employees. Perform data transformation for birthdate to age and also salary which is in rupees to salary in dollars using functions.

```
from datetime import datetime
def calculate_age(birthdate):
    today = datetime.now()
    birthdate = datetime.strptime(birthdate, "%Y-%m-%d")
    return today.year - birthdate.year - ((today.month, today.day)
<(birthdate.month, birthdate.day))
def salary_in_dollars(salary_in_rupees, conversion_rate=82.5):
    return salary_in_rupees / conversion_rate
birthdate = input("ENTER BIRTHDATE (YYYY-MM-DD):")
salary = float(input("ENTER SALARY IN RUPEES; "))
age = calculate_age(birthdate)
salary_usd = salary_in_dollars(salary)
print(f"AGE: {age} years")
print(f"SALARY IN USD: ${salary_usd:.2f}")
```

```
>>>
    ================== RESTART: C:/Users/Student/Desktop/IT/3.py ============
    ENTER BIRTHDATE (YYYY-MM-DD):2007-11-21
    ENTER SALARY IN RUPEES; \10000000
    AGE: 18 years
    SALARY IN USD: $121212.12
>>>
```

IV] Print the reverse number of a given number

```
number = int(input("Enter a number: "))
reverse_number = int(str(number)[::-1])
print(f"REVERSED NUMBER: {reverse_number}")
```

```
========================================================================== RESTART: C:/Users/Student/Desktop/IT/4.py ==========================================
Enter a number: 123456789
REVERSED NUMBER: 987654321
>>>
```

ALL

1.py - C:/Users/Student/Desktop/IT/1.py (3.13.1)

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```python
# Program to calculate momentum
mass = float(input("Enter mass in kg: "))
velocity = float(input("Enter velocity in m/s: "))
momentum = mass * velocity
print("Momentum =", momentum, "kg m/s")
```

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2.py - C:/Users/Student/Desktop/IT/2.py (3.13.1)

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```python
import math
n = int(input("ENTER A NUMBER:"))
if 0 <= n<10:
   print(f"Square of {n}: {n**2}")
elif 10 <= n <100:
   print(f"Square root of {n}: {math.sqrt(n):.2f}")
elif 100 <= n <1000:
   print(f"Cube root of {n}: {n**(1/3):.2f}")
else:
   print("PLEASE ENTER A NUMBER BETWEEN 0 AND 999. ")
```

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3.py - C:/Users/Student/Desktop/IT/3.py (3.13.1)

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```python
from datetime import datetime
def calculate_age(birthdate):
    today = datetime.now()
    birthdate = datetime.strptime(birthdate, "%Y-%m-%d")
    return today.year - birthdate.year - ((today.month, today.day) <(birthdate.month, birthdate.day))
def salary_in_dollars(salary_in_rupees, conversion_rate=82.5):
    return salary_in_rupees / conversion_rate
birthdate = input("ENTER BIRTHDATE (YYYY-MM-DD):")
salary = float(input("ENTER SALARY IN RUPEES: "))
age = calculate_age(birthdate)
salary_usd = salary_in_dollars(salary)
print(f"AGE: {age} years")
print(f"SALARY IN USD: ${salary_usd:.2f}")
```

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4.py - C:/Users/Student/Desktop/IT/4.py (3.13.1)

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```python
number = int(input("Enter a number: "))
reverse_number = int(str(number)[::-1])
print(f"REVERSED NUMBER: {reverse_number}")
```

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IDLE Shell 3.13.1

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```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: C:/Users/Student/Desktop/IT/1.py ==========
Enter mass in kg: 70
Enter velocity in m/s: 60
Momentum = 4200.0 kg m/s
>>>
========== RESTART: C:/Users/Student/Desktop/IT/2.py ==========
ENTER A NUMBER:400
Cube root of 400: 7.37
>>>
========== RESTART: C:/Users/Student/Desktop/IT/3.py ==========
ENTER BIRTHDATE (YYYY-MM-DD):2007-11-21
ENTER SALARY IN RUPEES: \10000000
AGE: 18 years
SALARY IN USD: $121212.12
>>>
========================================================================== RESTART: C:/Users/Student/Desktop/IT/4.py ==========================================================================
Enter a number: 123456789
REVERSED NUMBER: 987654321
>>>
```

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